

Chapter 8: Working with Fractions

Worksheet 8: Working with Fractions

Learning Objective: Add, subtract, multiply and divide fractions and mixed numbers, and apply these operations to real-life recipe and measurement contexts.

Worked Example:

A recipe needs $2\frac{1}{2}$ cups of flour for one batch. How much flour is needed for 3 batches?

Solution: $2\frac{1}{2} = \frac{5}{2}$. $\frac{5}{2} \times 3 = \frac{15}{2} = 7\frac{1}{2}$ cups.

Questions (1–20)

1. Add: $\frac{2}{5} + \frac{3}{10}$.

2. Subtract: $\frac{7}{8} - \frac{1}{4}$.

3. Multiply: $\frac{2}{3} \times \frac{4}{5}$.

4. Divide: $\frac{3}{4} \div \frac{2}{3}$.

5. Convert $\frac{17}{5}$ to a mixed number.

6. Add: $1\frac{1}{2} + 2\frac{2}{3}$.

7. A water tank is $\frac{3}{4}$ full. If $\frac{1}{8}$ of its capacity is used, what fraction remains?

-
8. Multiply: $1\frac{1}{4} \times \frac{2}{5}$.
9. Find the reciprocal of $\frac{5}{8}$.
10. Divide: $4 \div \frac{2}{3}$.
11. A tailor has $8\frac{1}{2}$ m of cloth and uses $\frac{2}{3}$ of it for one dress. How much cloth is used?
How much remains?
12. A recipe requires $\frac{3}{4}$ cup sugar. How much sugar is needed for $2\frac{1}{2}$ times the recipe?
13. Simplify: $(\frac{2}{3} + \frac{1}{6}) \div \frac{1}{2}$.
14. A pizza is cut into 8 slices. Ravi eats $\frac{3}{8}$ and Meena eats $\frac{1}{4}$. What fraction is left?
15. A rope of length $6\frac{3}{4}$ m is cut into pieces of $\frac{3}{4}$ m each. How many pieces can be cut?

-
16. A worker completes $\frac{2}{5}$ of a job in one day. How many days will the whole job take, and how much is left after 2 days?
17. A tank holds 18 litres when it is $\frac{3}{4}$ full. Find its full capacity.
18. A car uses $\frac{3}{5}$ of a tank of fuel for a trip of 240 km. How far can it travel on a full tank at the same rate?
19. Look at a recipe at home using fractions (cups, spoons). If you double the recipe, recalculate the fraction amounts for two ingredients.
20. If you spend $\frac{2}{5}$ of your day sleeping and $\frac{1}{4}$ at school, what fraction of the day is spent on other activities? Convert this to hours.

Reflection Question: Where in your daily life (cooking, sharing, time) do you use fractions without realising it? Write one example.

Real-Life Application Question: Find a packaged food label or recipe that lists a fraction (like $\frac{1}{2}$ cup or $\frac{3}{4}$ spoon). Write it down and explain what it means.